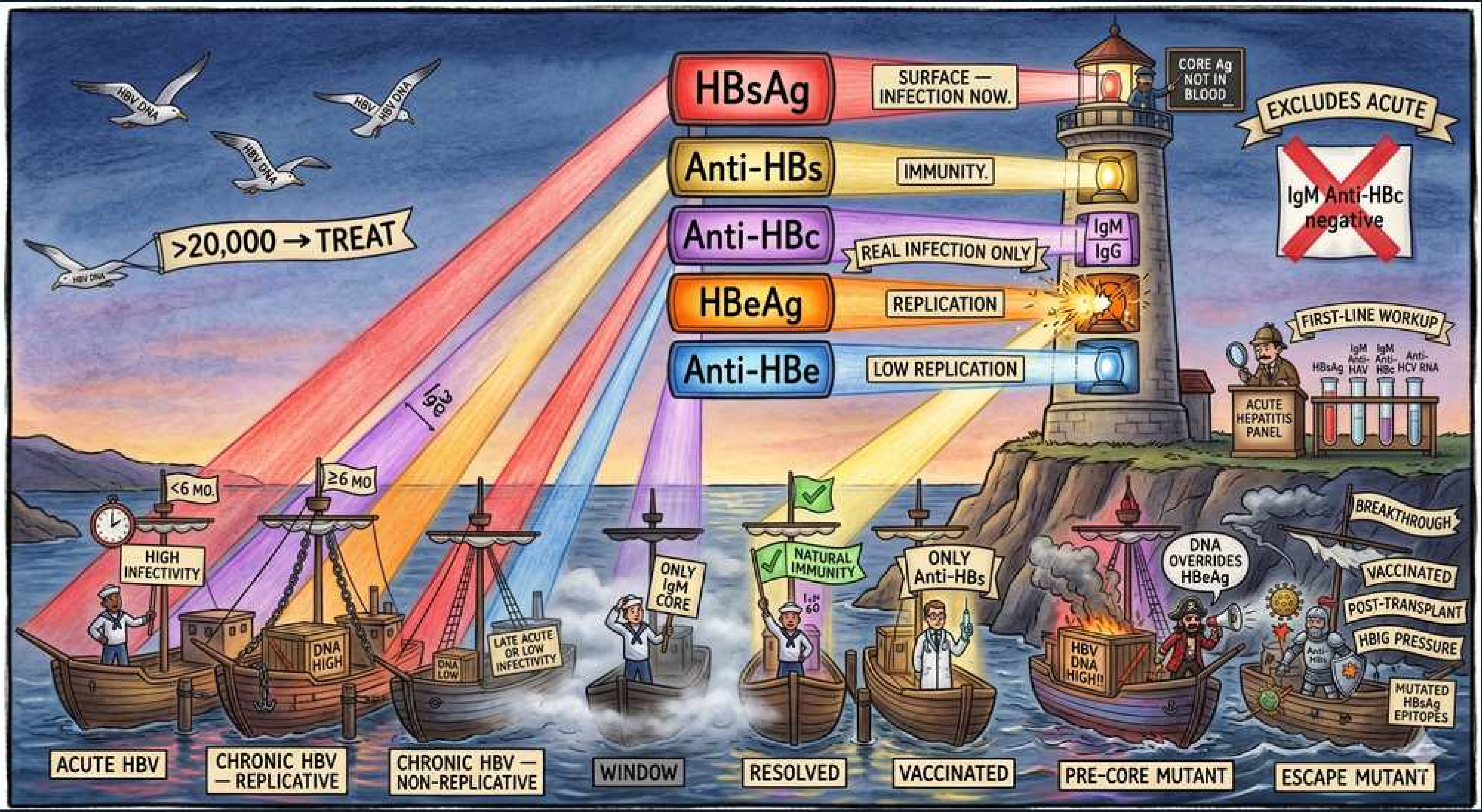
Hepatitis B — Serologic Marker Lighthouse (Interpretation)



1. HBsAg = infection now

WHAT YOU SEE

Red banner 'HBsAg — SURFACE — INFECTION NOW'

WHAT IT ENCODES

HBsAg (surface antigen) is the first marker to appear (~1–10 weeks after exposure) and means ACTIVE infection — acute or chronic — and infectivity. Persistence of HBsAg for >6 months defines CHRONIC hepatitis B. It is also the antigen used in the vaccine, so vaccination produces anti-HBs but NEVER HBsAg positivity.

BOARD TRAP

HBsAg positive = current infection (and contagious), regardless of symptoms. A vaccinated person is HBsAg-NEGATIVE / anti-HBs-positive — vaccination never makes HBsAg positive.

---

2. Anti-HBs = immunity

WHAT YOU SEE

'Anti-HBs — IMMUNITY' banner

WHAT IT ENCODES

Anti-HBs (surface antibody) is the PROTECTIVE, neutralizing antibody. Its presence means immunity — either from resolved natural infection (anti-HBs + anti-HBc both positive) or from VACCINATION (anti-HBs positive, anti-HBc NEGATIVE). A titer ≥10 mIU/mL is considered protective.

BOARD TRAP

Anti-HBs distinguishes immune from infected. Anti-HBs alone (no anti-HBc) = vaccinated; anti-HBs WITH anti-HBc = resolved natural infection. Don't call a vaccinated patient 'previously infected.'

---

3. Anti-HBc = real infection only

WHAT YOU SEE

Purple 'Anti-HBc — REAL INFECTION ONLY', IgM/IgG split

WHAT IT ENCODES

Total anti-HBc appears with any natural infection and persists for life — it is NEVER produced by vaccination, so it is the marker that proves true exposure. IgM anti-HBc indicates ACUTE/recent infection and is the key marker in the window period; IgG anti-HBc indicates remote/chronic infection. Isolated anti-HBc (no HBsAg, no anti-HBs) can mean window period, resolved infection with waned anti-HBs, or occult HBV.

BOARD TRAP

Anti-HBc positivity = the patient was truly infected at some point — a vaccinated person will be anti-HBc NEGATIVE. IgM anti-HBc is what nails ACUTE HBV during the window when HBsAg may already be gone.

---

4. HBeAg = high replication / infectivity

WHAT YOU SEE

Orange 'HBeAg — REPLICATION', flames

WHAT IT ENCODES

HBeAg reflects active viral replication and HIGH infectivity (high HBV DNA). HBeAg-positive chronic HBV is the immune-active, highly transmissible phase. HBeAg-positive pregnant mothers have the greatest perinatal transmission risk. Seroconversion from HBeAg to anti-HBe usually signals declining replication.

BOARD TRAP

HBeAg positive = highly infectious / high viral load — the opposite of anti-HBe. Don't equate HBeAg with HBsAg; HBeAg is a replication marker, HBsAg is the infection/chronicity marker.

---

5. Anti-HBe = low replication

WHAT YOU SEE

Blue 'Anti-HBe — LOW REPLICATION'

WHAT IT ENCODES

Anti-HBe appears after HBeAg seroconversion and generally indicates LOW replication / lower infectivity — the inactive carrier state. Exception: pre-core MUTANT HBV can have anti-HBe positive yet high HBV DNA and active disease, so anti-HBe does not always equal quiescence.

BOARD TRAP

Anti-HBe usually means low replication, BUT pre-core/basal-core-promoter mutants are anti-HBe positive with HIGH DNA and active hepatitis — check HBV DNA, don't rely on anti-HBe alone.

---

6. IgM anti-HBc excludes/confirms acute

WHAT YOU SEE

Top-right plaque 'EXCLUDES ACUTE — IgM Anti-HBc NEGATIVE'

WHAT IT ENCODES

IgM anti-HBc is the discriminator for ACUTE versus chronic HBV and is essential during the window period. A NEGATIVE IgM anti-HBc effectively excludes acute infection (points to chronic if HBsAg+). A POSITIVE IgM anti-HBc with HBsAg+ confirms acute hepatitis B.

BOARD TRAP

To tell acute from chronic HBV when HBsAg is positive, order IgM anti-HBc — it is positive in acute and negative in chronic. HBsAg alone cannot make that distinction.

---

7. Acute HBV ship (<6 mo)

WHAT YOU SEE

Far-left ship, clock '<6 MO', 'HIGH INFECTIVITY'

WHAT IT ENCODES

Acute HBV (<6 months): HBsAg+, HBeAg+ (high infectivity), IgM anti-HBc+, anti-HBs negative. Most healthy adults clear it (>95%), losing HBsAg and developing anti-HBs. The <6-month clock is the threshold; persistence beyond it defines chronicity.

BOARD TRAP

Adult-acquired acute HBV usually resolves (~95%); it is NEONATAL/early-childhood infection that becomes chronic ~90% of the time. Don't assume acute adult HBV will become chronic.

---

8. Chronic HBV — replicative

WHAT YOU SEE

Ship '≥6 MO', HBeAg flame / high DNA

WHAT IT ENCODES

Chronic replicative HBV: HBsAg+ for ≥6 months, HBeAg+, high HBV DNA, IgG anti-HBc+, anti-HBs negative. This immune-active phase drives liver injury, fibrosis, and HCC risk and is the phase usually treated (with elevated ALT/fibrosis).

BOARD TRAP

HBsAg persisting ≥6 months = chronic — the dividing line from acute. Replicative (HBeAg+/high DNA) vs non-replicative is determined by HBeAg and DNA, not by HBsAg alone.

---

9. Chronic HBV — non-replicative

WHAT YOU SEE

Ship 'HBeAg seroconverted, low DNA'

WHAT IT ENCODES

Inactive carrier / non-replicative chronic HBV: HBsAg+, HBeAg NEGATIVE, anti-HBe+, low/undetectable HBV DNA, normal ALT, IgG anti-HBc+. Lower infectivity and low short-term progression risk, but requires monitoring (reactivation, pre-core mutant, HCC surveillance) and never assume permanent quiescence.

BOARD TRAP

An inactive carrier is still HBsAg-positive and can reactivate — especially with immunosuppression/chemotherapy. Give antiviral prophylaxis before immunosuppression even in inactive carriers.

---

10. Window period

WHAT YOU SEE

'WINDOW — ONLY anti-HBc' ship, sailor with anti-HBc only

WHAT IT ENCODES

The window period: HBsAg has cleared but anti-HBs has not yet appeared — the ONLY positive marker is anti-HBc (IgM during acute window). Recognizing this prevents falsely calling the patient uninfected. Total anti-HBc is the bridge marker across this gap.

BOARD TRAP

In the window period, HBsAg is negative and anti-HBs is negative — without checking anti-HBc you'd wrongly conclude 'no HBV.' Isolated anti-HBc demands IgM testing and HBV DNA.

---

11. Resolved infection

WHAT YOU SEE

Ship 'NATURAL — anti-HBs + anti-HBc', green check

WHAT IT ENCODES

Resolved past HBV: HBsAg negative, anti-HBs POSITIVE, anti-HBc (IgG) POSITIVE. Immune by natural infection. The presence of anti-HBc is what separates this from a vaccinated person.

BOARD TRAP

Resolved natural infection = anti-HBs+ AND anti-HBc+. Vaccinated = anti-HBs+ but anti-HBc NEGATIVE. The anti-HBc is the single tie-breaker between the two immune states.

---

12. Vaccinated

WHAT YOU SEE

Ship 'ONLY anti-HBs', vaccine vial

WHAT IT ENCODES

Vaccine-induced immunity: anti-HBs POSITIVE, with HBsAg, HBeAg, and anti-HBc all NEGATIVE. Because the vaccine contains only recombinant HBsAg, the body makes anti-HBs but never anti-HBc. Protective titer ≥10 mIU/mL.

BOARD TRAP

Isolated anti-HBs (everything else negative) = vaccinated, NOT infection. If anti-HBc were also positive, it would be resolved natural infection instead.

---

13. Pre-core mutant

WHAT YOU SEE

Ship 'DNA OVERRIDES HBeAg', flames despite anti-HBe

WHAT IT ENCODES

Pre-core / basal-core-promoter MUTANT HBV cannot produce HBeAg, so serology shows HBeAg NEGATIVE / anti-HBe POSITIVE yet HBV DNA is high and ALT elevated with active necroinflammation. More common in Mediterranean/Asian strains; runs a more aggressive, fluctuating course.

BOARD TRAP

HBeAg-negative does NOT always mean inactive — pre-core mutant has anti-HBe+ but high DNA and active disease. Always confirm with HBV DNA, which 'overrides' the reassuring HBeAg result.

---

14. Escape mutant

WHAT YOU SEE

Far-right 'ESCAPE MUTANT', mutated HBsAg epitope

WHAT IT ENCODES

S-gene 'escape' mutants alter the HBsAg epitope so that vaccine-induced or HBIG-induced anti-HBs no longer neutralizes the virus — causing breakthrough infection, vaccine failure, or post-transplant/HBIG-pressure reactivation. Explains rare HBsAg-positive disease in vaccinated or HBIG-treated patients.

BOARD TRAP

A vaccinated/anti-HBs-positive patient who still gets HBV may harbor an escape mutant — vaccine protection is not absolute. This is distinct from waning titer and from occult HBV.

---

15. >20,000 → treat

WHAT YOU SEE

Left banner '>20,000 → TREAT'

WHAT IT ENCODES

Treatment threshold anchor: HBeAg-positive chronic HBV with HBV DNA >20,000 IU/mL plus elevated ALT and/or significant fibrosis warrants antiviral therapy (entecavir or tenofovir first-line). HBeAg-negative disease uses a lower DNA threshold (>2,000 IU/mL). All cirrhotics with detectable DNA are treated.

BOARD TRAP

Don't treat on DNA level alone — the immune-tolerant patient (very high DNA, normal ALT, no fibrosis) is monitored. Treatment needs the combination of DNA + ALT + fibrosis/cirrhosis.

---

16. First-line workup panel

WHAT YOU SEE

'FIRST-LINE WORKUP' sign: HBsAg, anti-HBs, anti-HBc + DNA, ALT, HCV/HDV

WHAT IT ENCODES

Initial chronic-HBV workup: the three serologies (HBsAg, anti-HBs, anti-HBc), then HBeAg/anti-HBe, HBV DNA, ALT, and fibrosis assessment, plus co-infection screening for HCV, HDV (in HBsAg+), and HIV. HCC surveillance with ultrasound ± AFP in at-risk groups (cirrhosis, certain ethnicities/ages, family history).

BOARD TRAP

Every newly diagnosed HBsAg-positive patient also needs HDV and HIV screening — HDV only co-infects/superinfects HBV carriers and dramatically worsens prognosis. Don't stop the workup at the basic HBV panel.
